

INTER IIT TECH MEET 13.0

High-resolution elemental mapping of
lunar surface

HIGH PREP 79

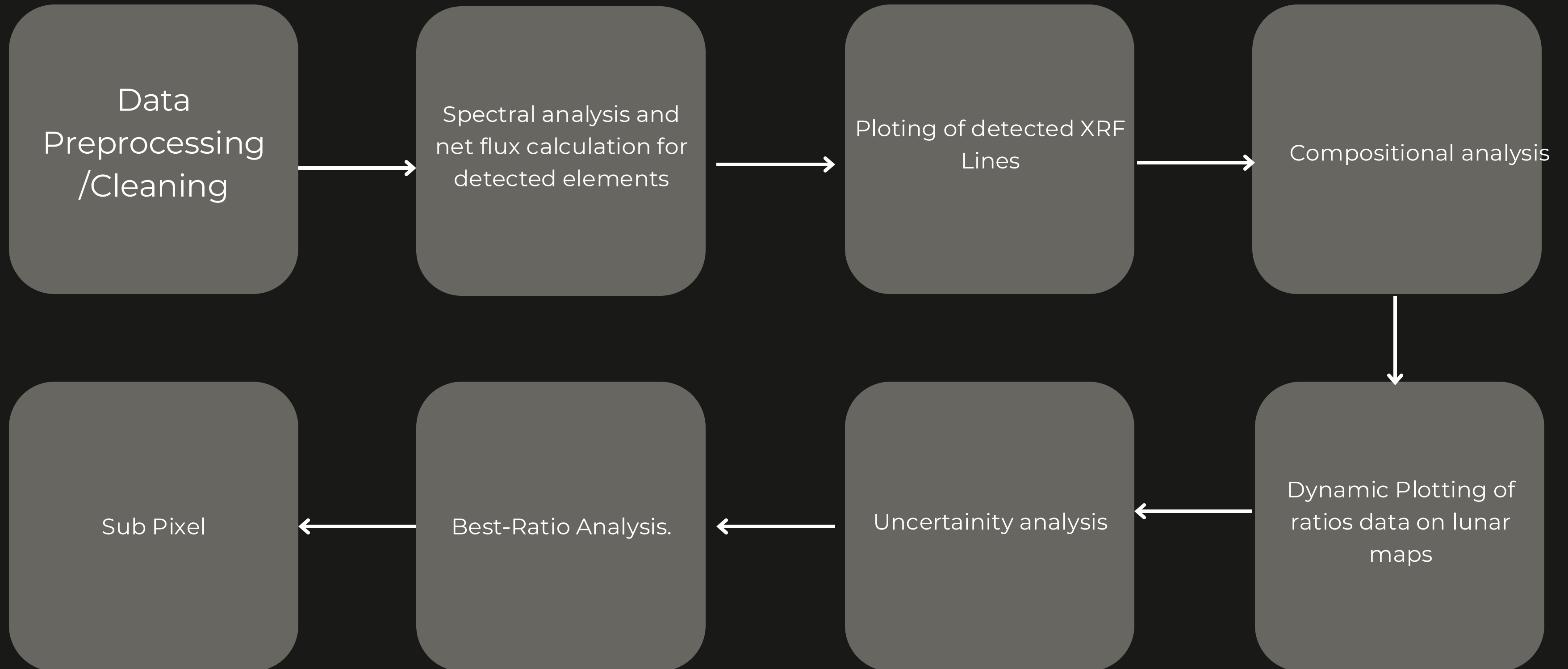


Objectives

- To create high-resolution, dynamic, and interactive elemental ratio maps that identify compositional variations at the kilometer scale, we propose the following approach:
 - Utilize XRF (X-ray Fluorescence) data to determine elemental detection and abundance ratios, which provide insights into compositional heterogeneity.
 - Map these elemental ratios onto the Lunar Reconnaissance Orbiter Wide Angle Camera (LROWAC) global map, ensuring precise alignment with lunar surface features.
 - Identify optimal elemental ratios that effectively represent compositional diversity within the terrain, capturing variations across different regions.
 - Employ overlapping map layers to achieve subpixel resolution, enhancing the level of detail and accuracy in identifying subtle compositional differences.



Approach, Outline of Steps Taken and Algorithm

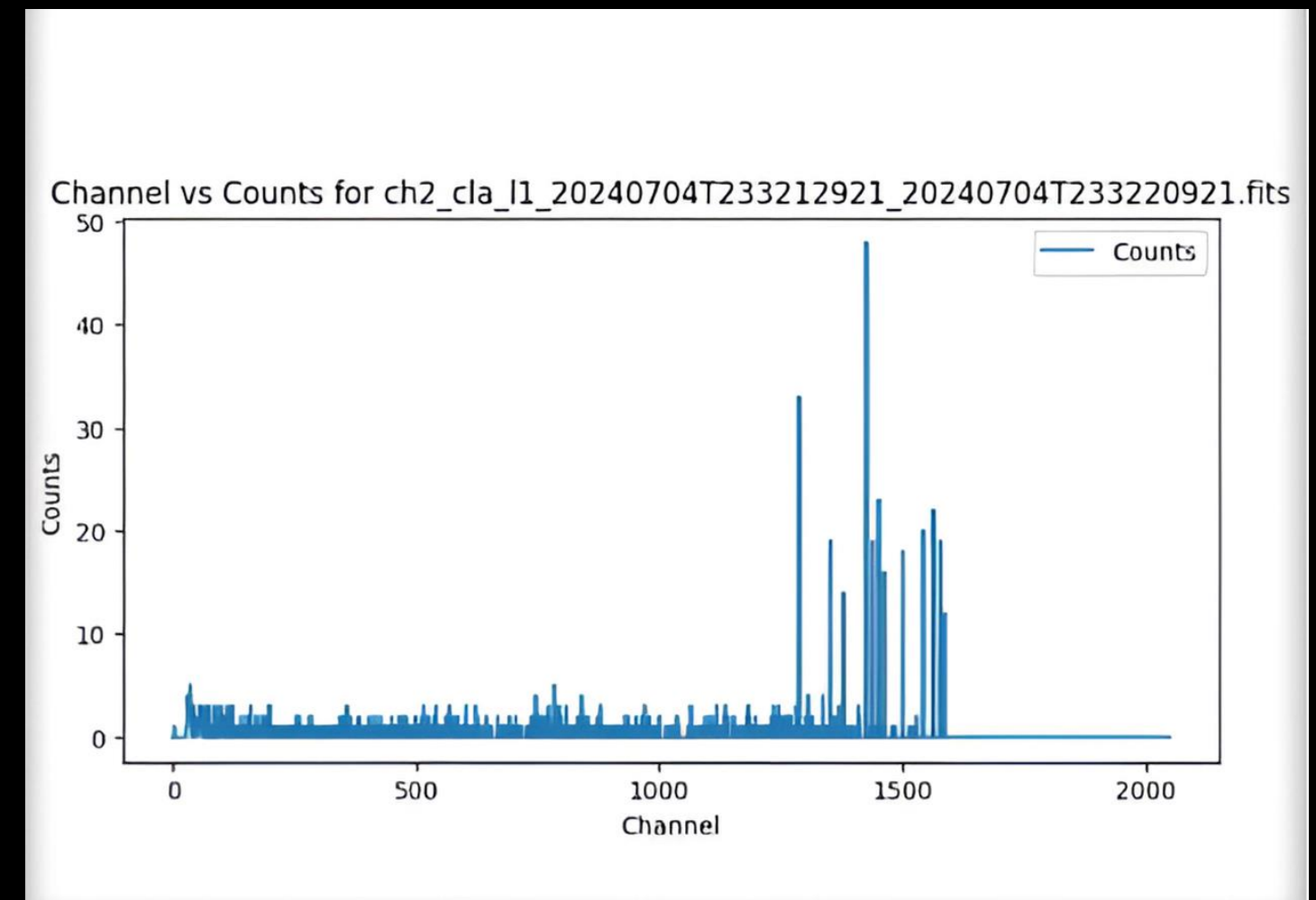


Data Preprocessing

The Data Set

The Chandrayaan 2's CLASS and XSM datasets were obtained from the ISRO's pradan website Chandrayaan 2 data repository. The CLASS data provides X-ray fluorescence (XRF) line intensities corresponding to the lunar surface, while the XSM data records the incident solar spectrum, which is useful for classifying high flare times.

- **Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS)** has FITS files and their respective XML file in PDS4 Format. which are used for spectral analysis of a particular area.
- **Solar X-Ray Monitor (XSM)** contains useable level 2 data which consists of light curves files and good time intervals for XSM device.



Sample FITS file

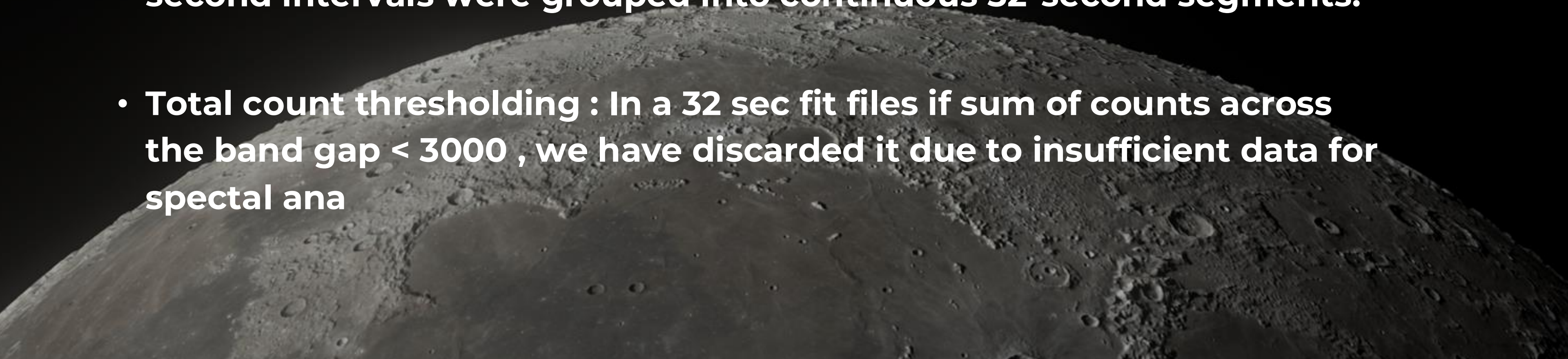
Data Preprocessing

- **Geotail : distortion of the spectrum of Al , Ca**

- Removal of time interval during the time when moon passes through geotail
- time intervals for the night side of the moon for geotail are extracted from the spk files
- Background noise is calculated for all night time data (solar angle > 120)
- Thresholding is done , and all the data above the threshold are removed

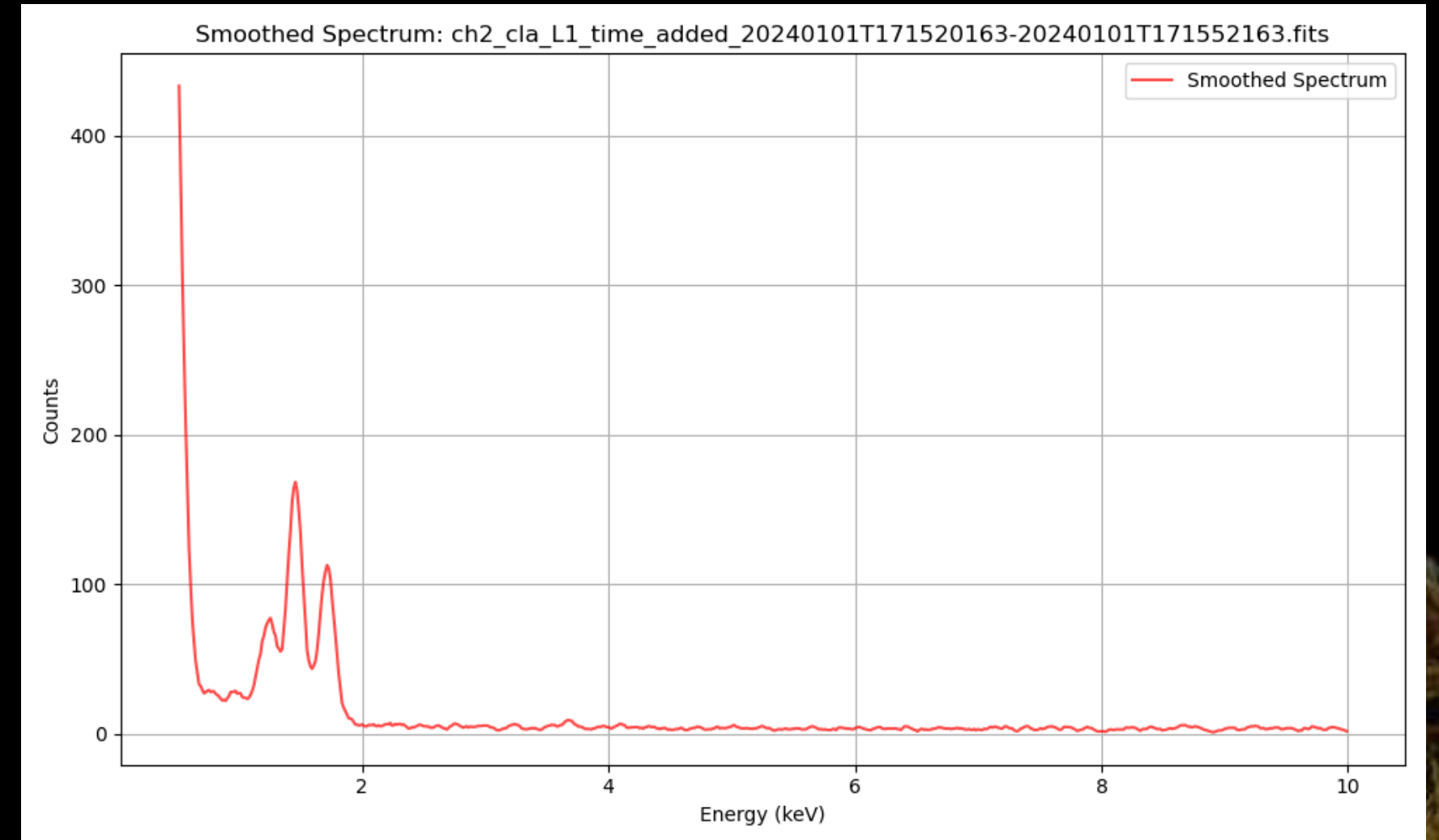
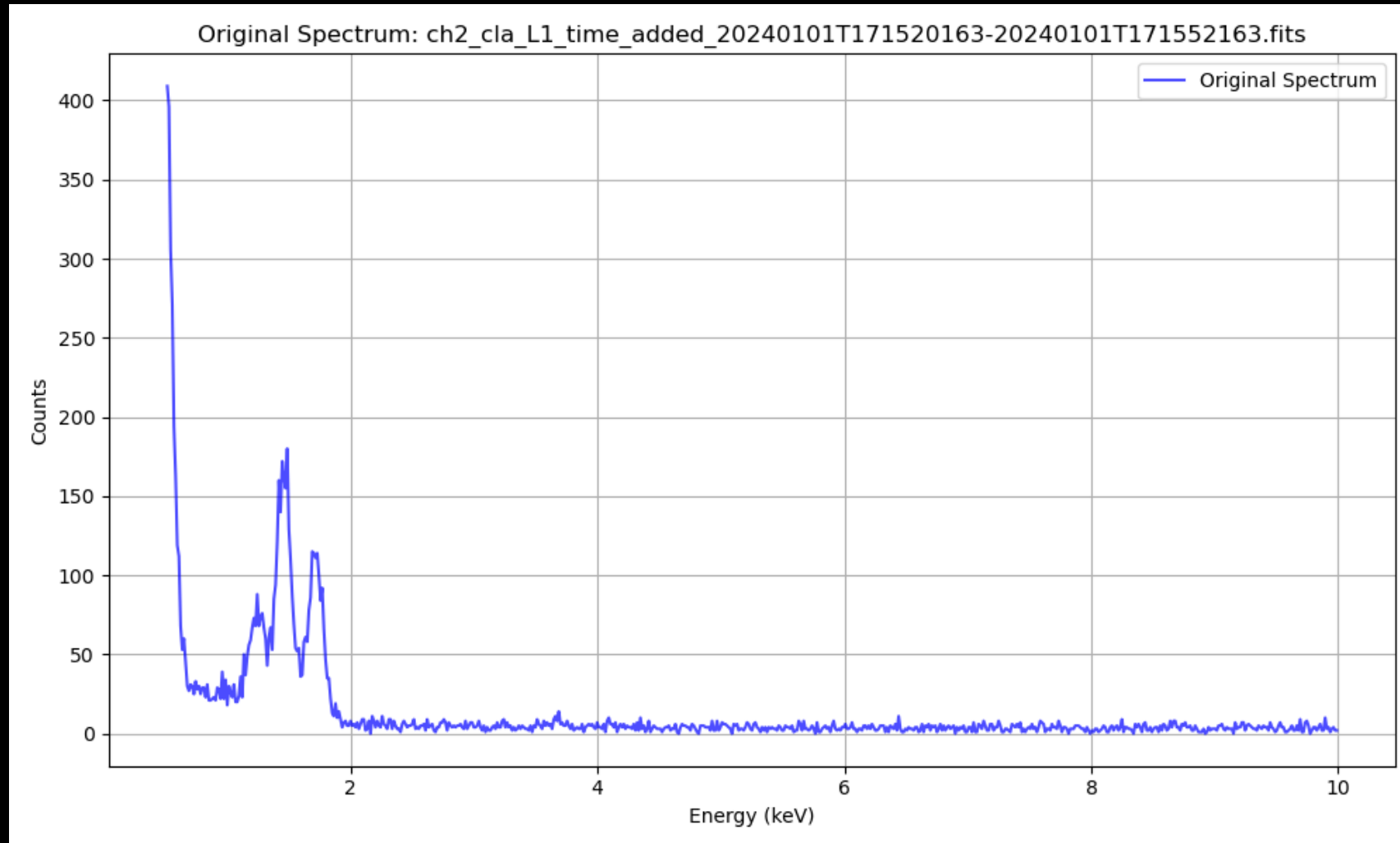
- **To meet the analysis requirement for longer durations, consecutive 8-second intervals were grouped into continuous 32-second segments.**

- **Total count thresholding : In a 32 sec fit files if sum of counts across the band gap < 3000 , we have discarded it due to insufficient data for spectral ana**

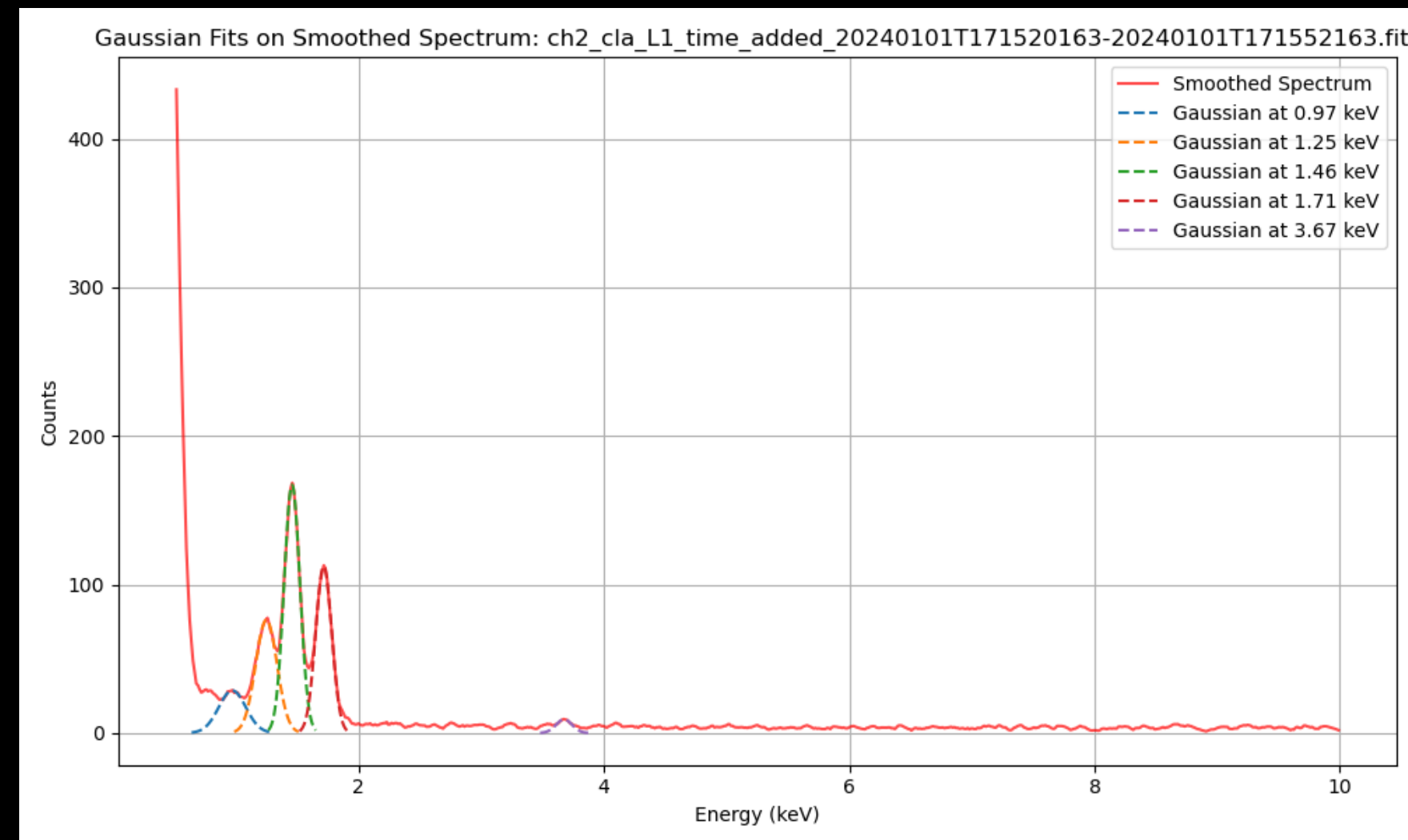


Spectral Analysis and Flux calculation

Smoothing: Smooth the counts data using the **Savitzky-Golay filter** to reduce noise while preserving peak features.



Peak detection and Gaussian fitting



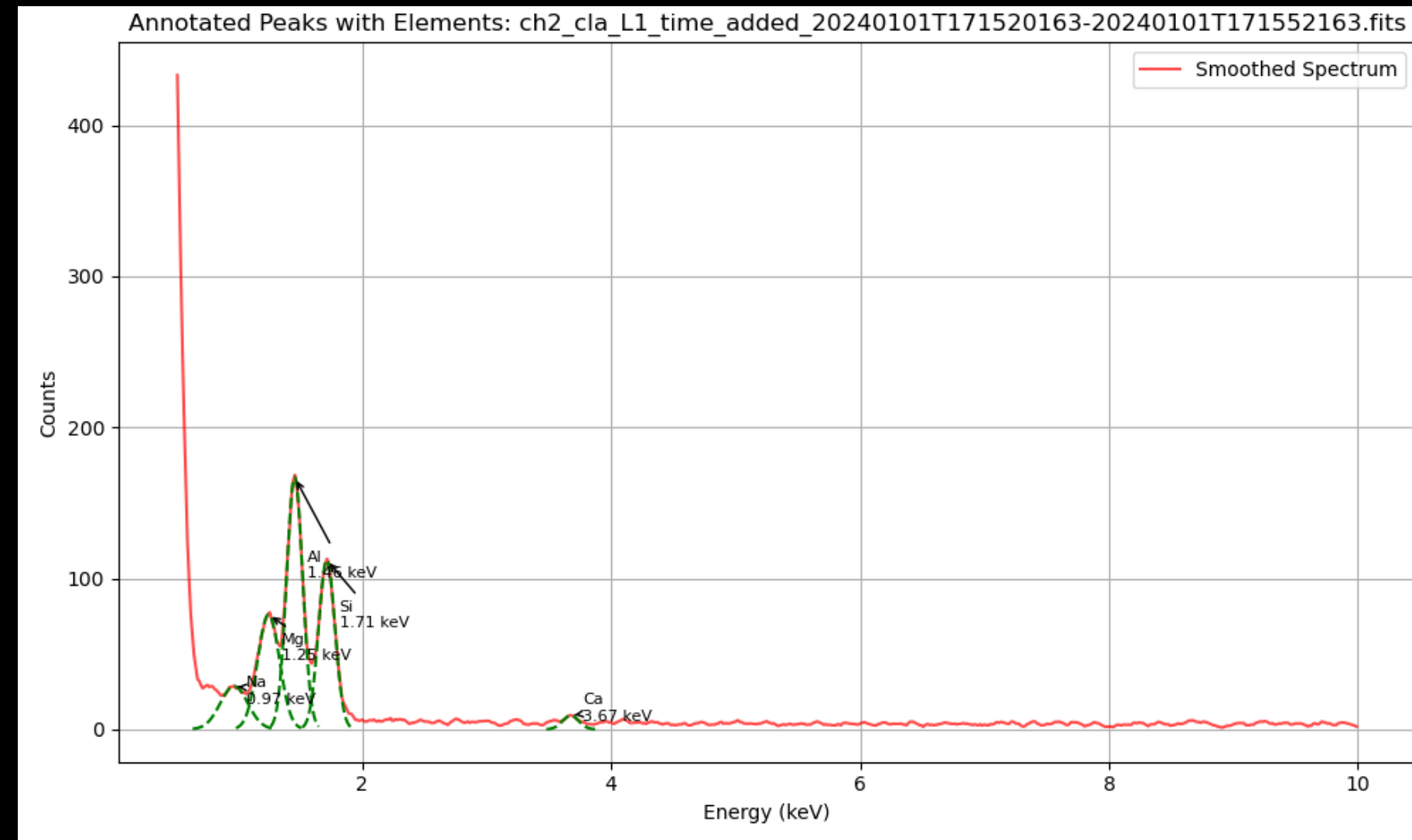
Peak Detection: Identify peaks using scipy's `signal.detect_peak()` function using prominence-based detection

Gaussian Curve Fitting: We fit a Gaussian function for analysis by extracting a small energy window (5 channels) around each detected peak.

$$G(x) = Ae^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where **A** represents the amplitude of the peak,
σ denotes the standard deviation of the Gaussian curve,
and **μ** is the mean of the Gaussian curve.

Peak detection and Gaussian fitting



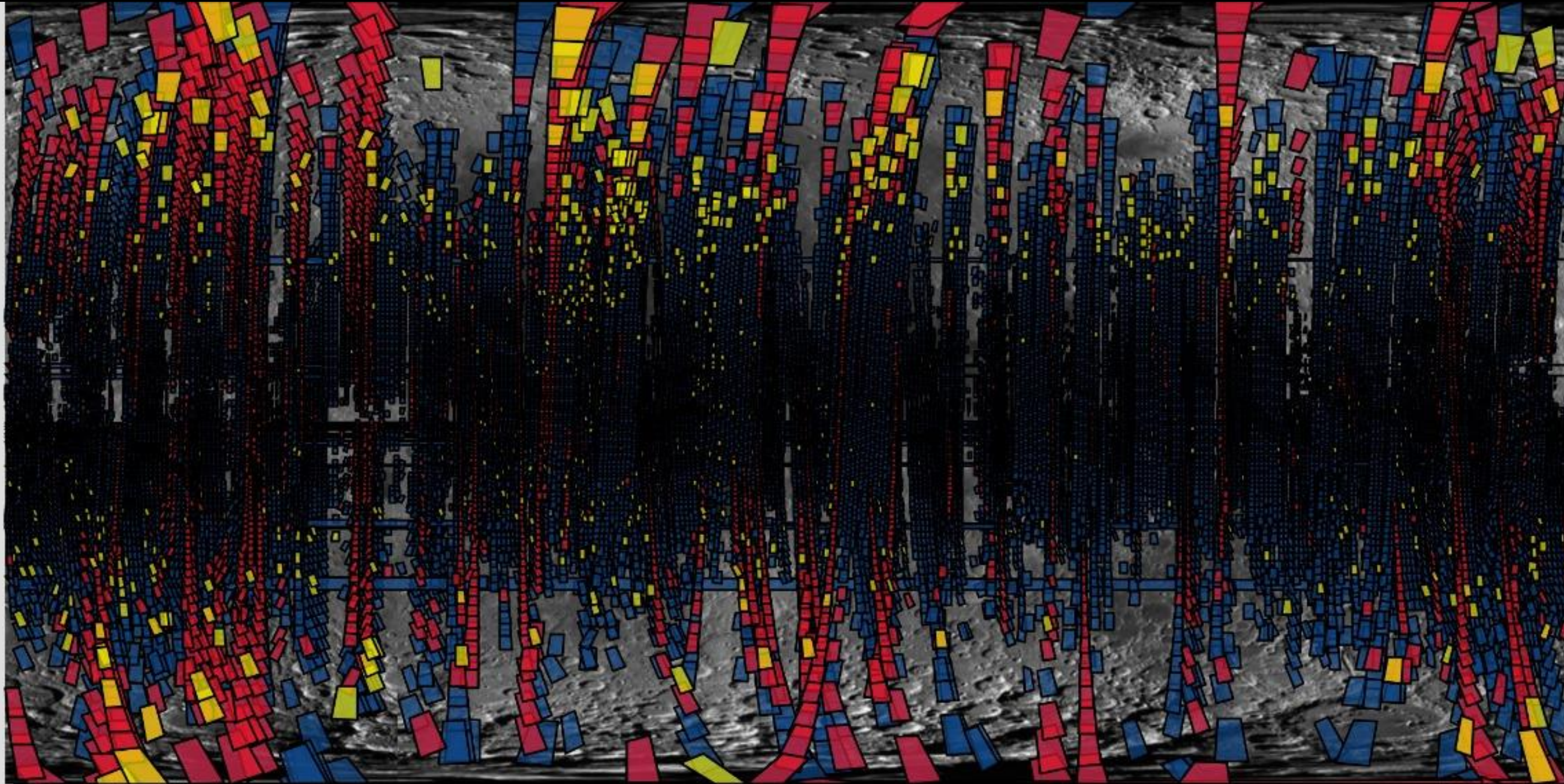
Element Identification: Match the fitted peak energies to known element emission lines ($K\alpha$, $K\beta$, $L\alpha$) within a tolerance range.

Gaussian Area and SNR Calculation: Compute the area under the Gaussian curve and the signal-to-noise ratio (SNR) for each peak.

$$Area = A\sqrt{2\pi}\sigma$$

$$SNR = A/\sigma$$

Detection of XRF line



Legend: Detected Elements

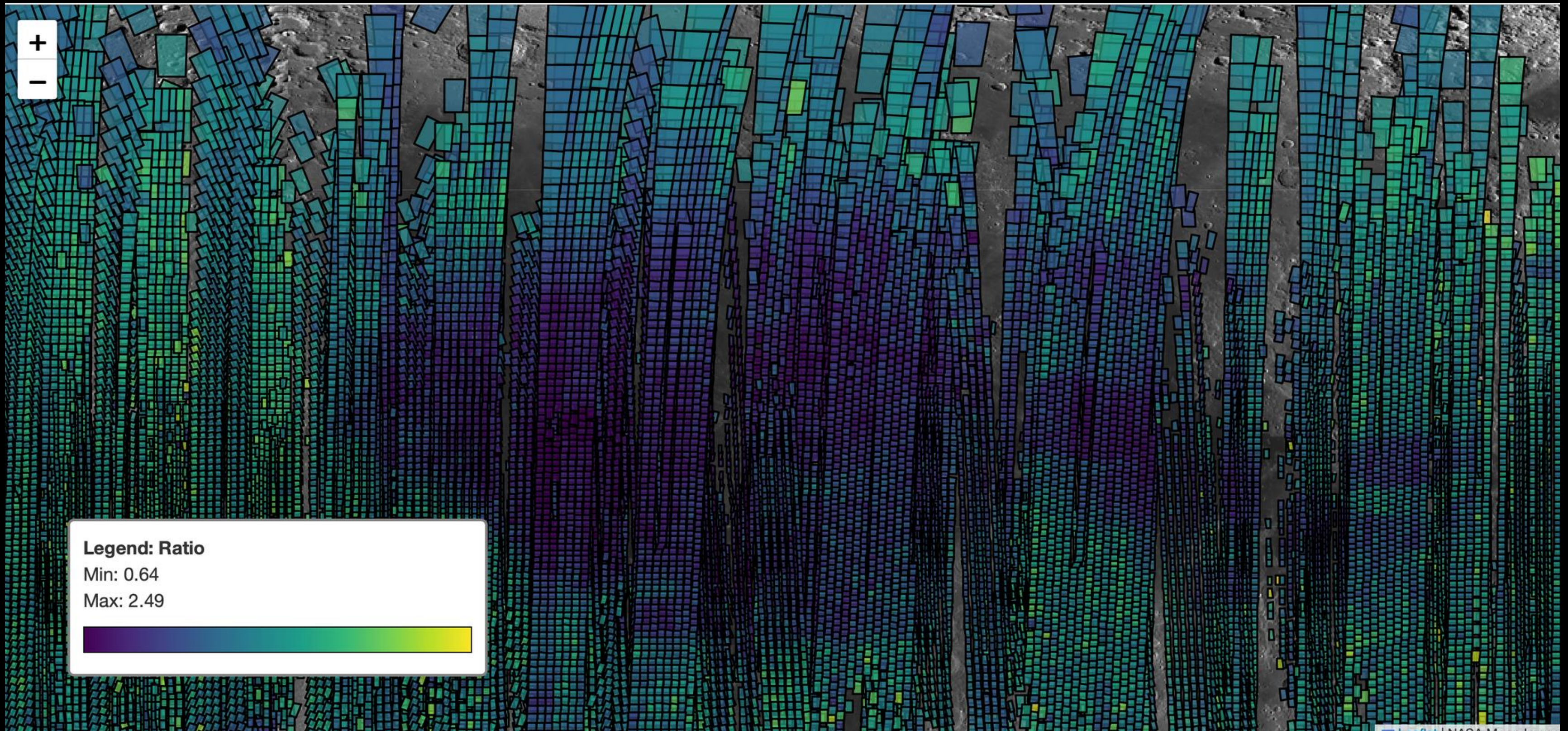
- Al
- Ca
- Na

Compositional Groups

- We can majorily divide the lunar surface in 3 types of terrain :
 - feldspathic highland
 - Mare basalts
 - Procellarum KREEP Terran
- **feldspathic highland** : anorthite (plagioclase feldspar): Nearly all the aluminum is in plagioclase, plagioclase is light colored.
- **Mare basalts** : Mainly pyroxene , Fe rich mineral , dark coloured
- ***Procellarum KREEP Terran***: contains less plagioclase and more pyroxene than in the feldspathic highlands.

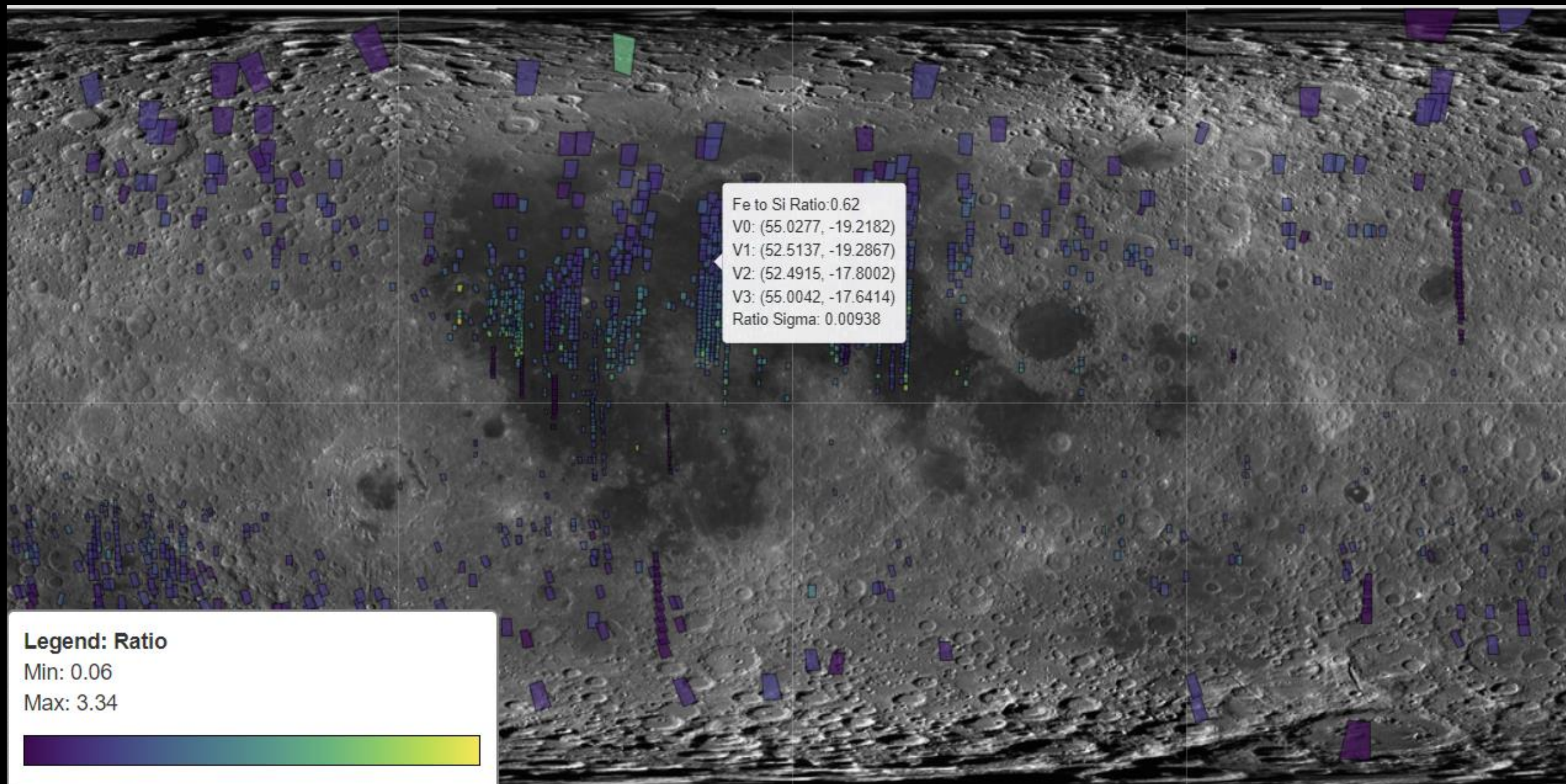


ALUMINIUM TO SILICON RATIO

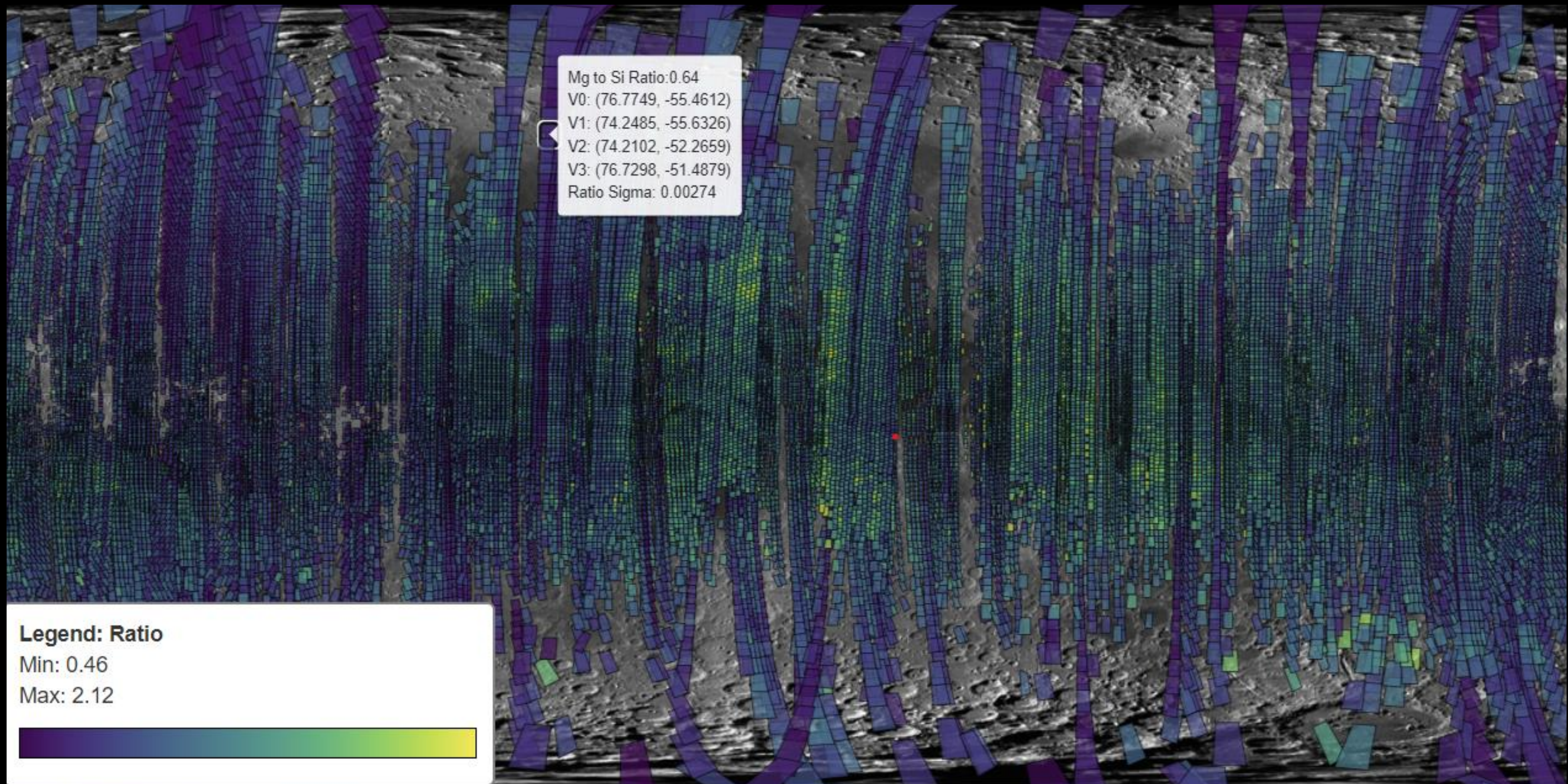


$\text{Al/Si} > 1.5$: ~ highland
 $\text{Al/Si} < 1$: Mare basalt
 $1 < \text{Al/Si} < 1.5$: KREEP

IRON TO SILICON RATIO



MAGNESIUM TO SILICON RATIO



Mg/Si < 1.1 : For high Lands and KREEP
Mg/Si > 1.1 : For Basalt

Uncertainty Analysis

$$G(x) = Ae^{-\frac{(x-\mu)^2}{\sigma^2 \cdot 2}}$$

$$R_g = \frac{A_1 \sigma_1}{A_2 \sigma_2} = \frac{A_1 \sqrt{2\pi} \cdot \sigma_1}{A_2 \sqrt{2\pi} \cdot \sigma_2}$$

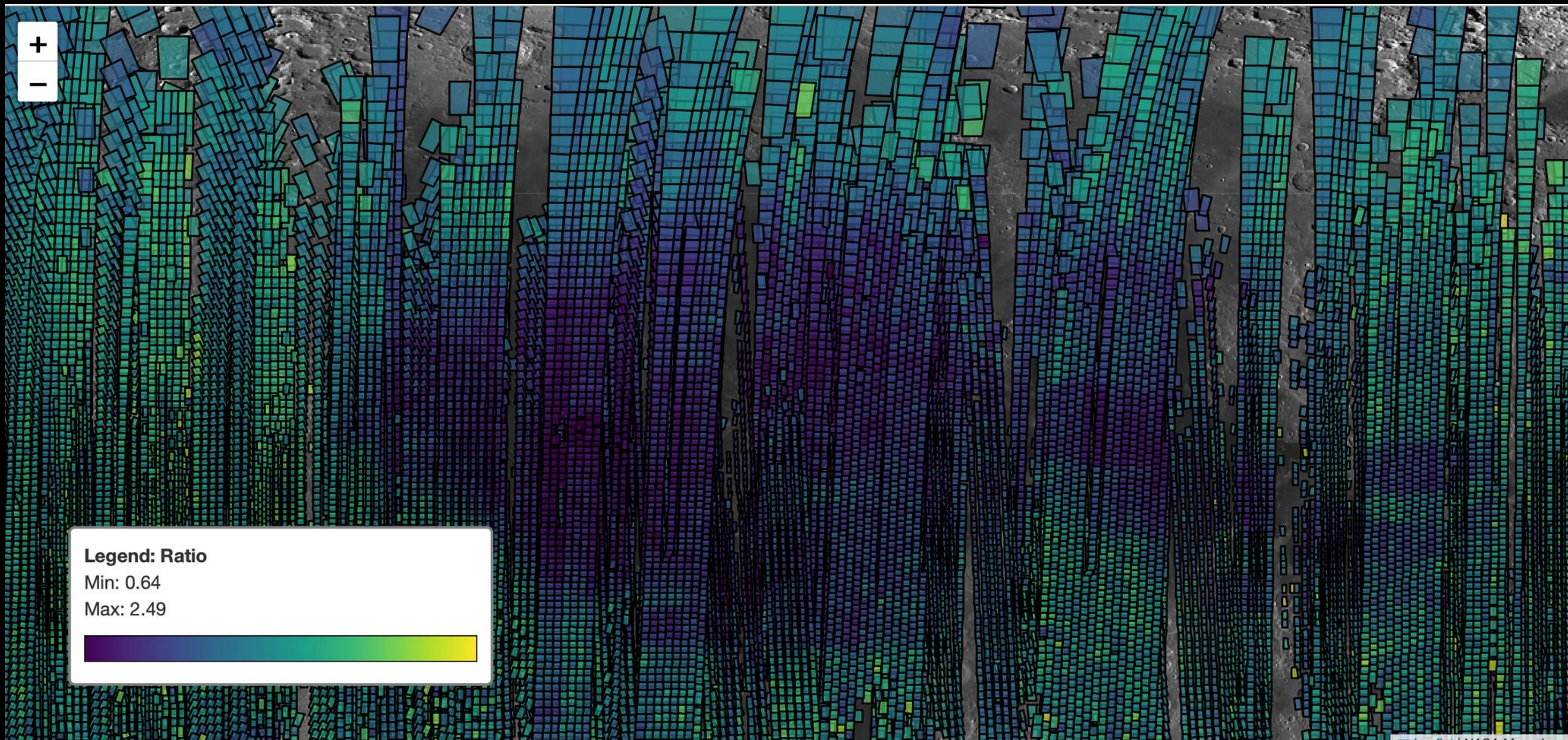
$$\frac{\delta R}{R} = \sqrt{\left(\frac{\delta A_1}{A_1}\right)^2 + \left(\frac{\delta \sigma_1}{\sigma_1}\right)^2 + \left(\frac{\delta A_2}{A_2}\right)^2 + \left(\frac{\delta \sigma_2}{\sigma_2}\right)^2}$$

$$\delta R = R \sqrt{\frac{2}{(\text{SNR}_1)^2} + \frac{2}{(\text{SNR}_2)^2}}$$

$$\text{Error}\% = 100 \cdot R \sqrt{\frac{2}{(\text{SNR}_1)^2} + \frac{2}{(\text{SNR}_2)^2}}$$

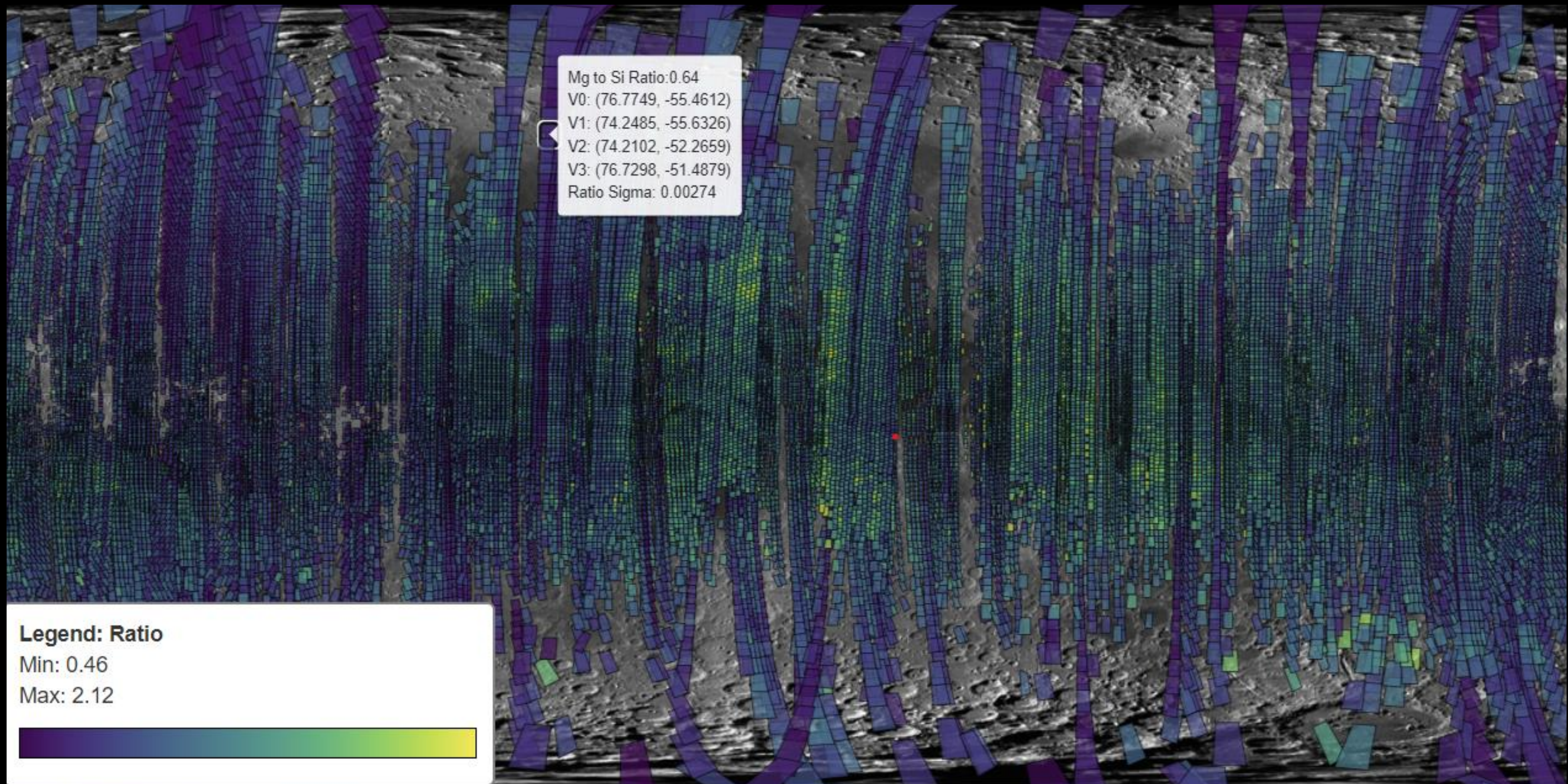
Ratio to Si Statistics per Element							
	Element	Mean	Median	Mode	Std_Dev	Minimum	Maximum
0	Al	1.413284	1.441773	1.500000	0.322187	0.554382	3.857558
1	Ca	0.150554	0.123673	0.086957	0.083628	0.061397	0.658124
2	Fe	0.879187	0.777083	1.000000	0.510196	0.062018	3.629032
3	Mg	1.027409	1.008114	1.000000	0.261924	0.350784	2.759230
4	Na	0.440956	0.424996	0.500000	0.126109	0.203679	1.988166
5	Si	1.000000	1.000000	1.000000	0.000000	1.000000	1.000000

As standard deviation of Al vs Si , Fe vs Si and Mg vs Si are high , these ratio will decipher the heterogeneity of surface of the moon



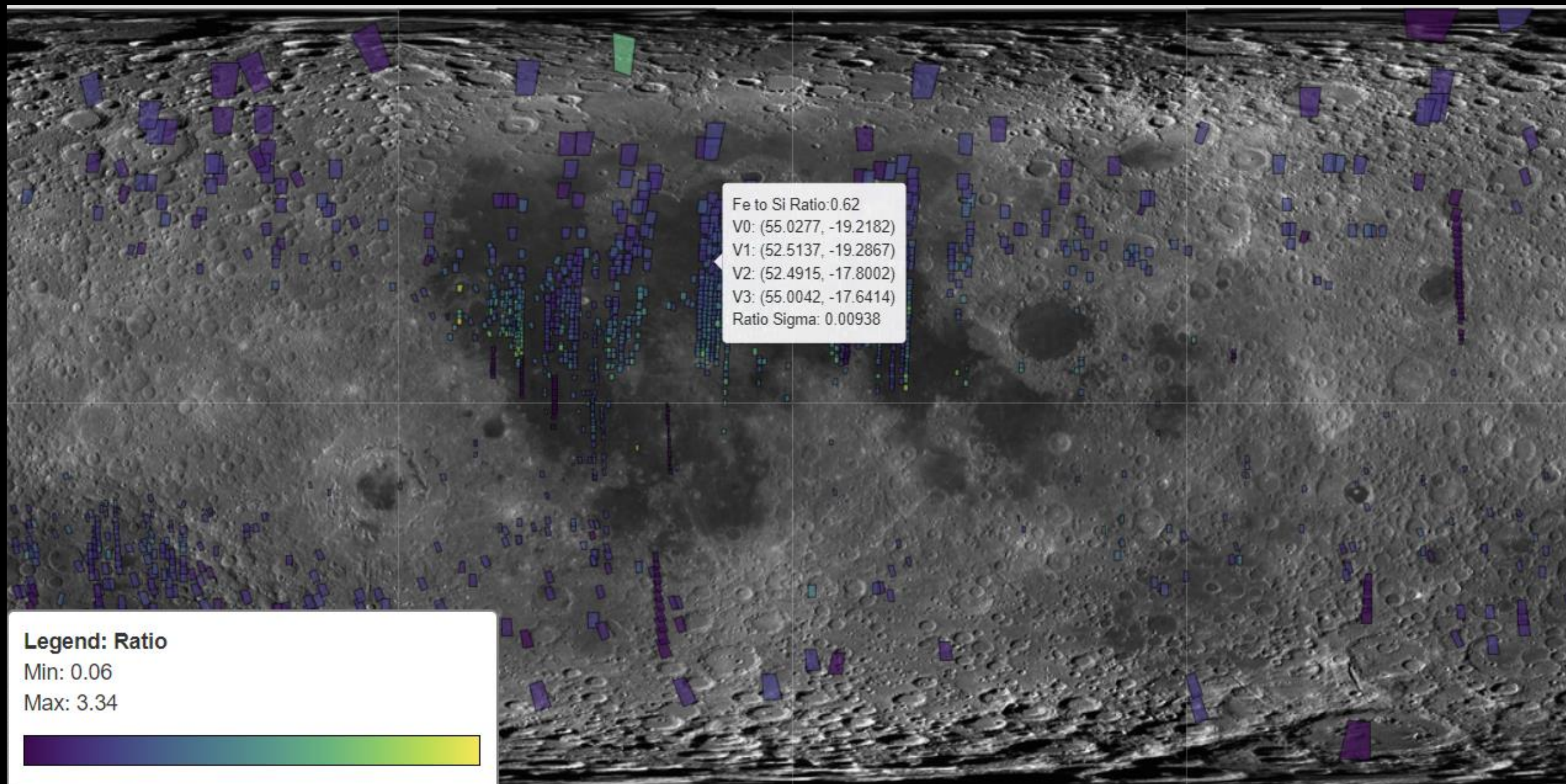
$\text{Al/Si} > 1$: ~ highland
 $\text{Al/Si} < 1$: Mare basalt

MAGNESIUM TO SILICON RATIO



Mg/Si < 1.1 : For high Lands and KREEP
Mg/Si > 1.1 : For Basalt

IRON TO SILICON RATIO



Live Demonstration



THANK YOU

Challenges and Lessons Learned

Challenges

- Geotail contamination impacting data quality.
- Filtering and aggregation while maintaining spatial and temporal integrity.
- Initially, we attempted to use QGIS for mapping and visualization of the lunar data. However, we faced challenges in converting the data into a dynamic 3D format suitable for interactive visualization and analysis. This necessitated the development of a custom software solution to address these limitations.

Lessons Learned

- Rigorous preprocessing ensures reliable datasets.
- Aggregation techniques maintain a balance between spatial resolution and statistical robustness.
- Developing a custom software solution allowed us to dynamically visualize lunar data and enabled seamless integration of 3D surface imagery with processed elemental data.

GEOTAIL

The geotail, a feature of Earth's magnetosphere, extends away from the Earth in the direction opposite to the Sun due to solar wind interactions. It is an elongated region populated with high-energy charged particles, which can significantly impact the quality of space-based X-ray observations, including those by Chandrayaan-2's. We are using SPICE data for finding out geotail intervals.



GOOD TIME INTERVAL

1. Identification of Good Time Intervals (GTI):

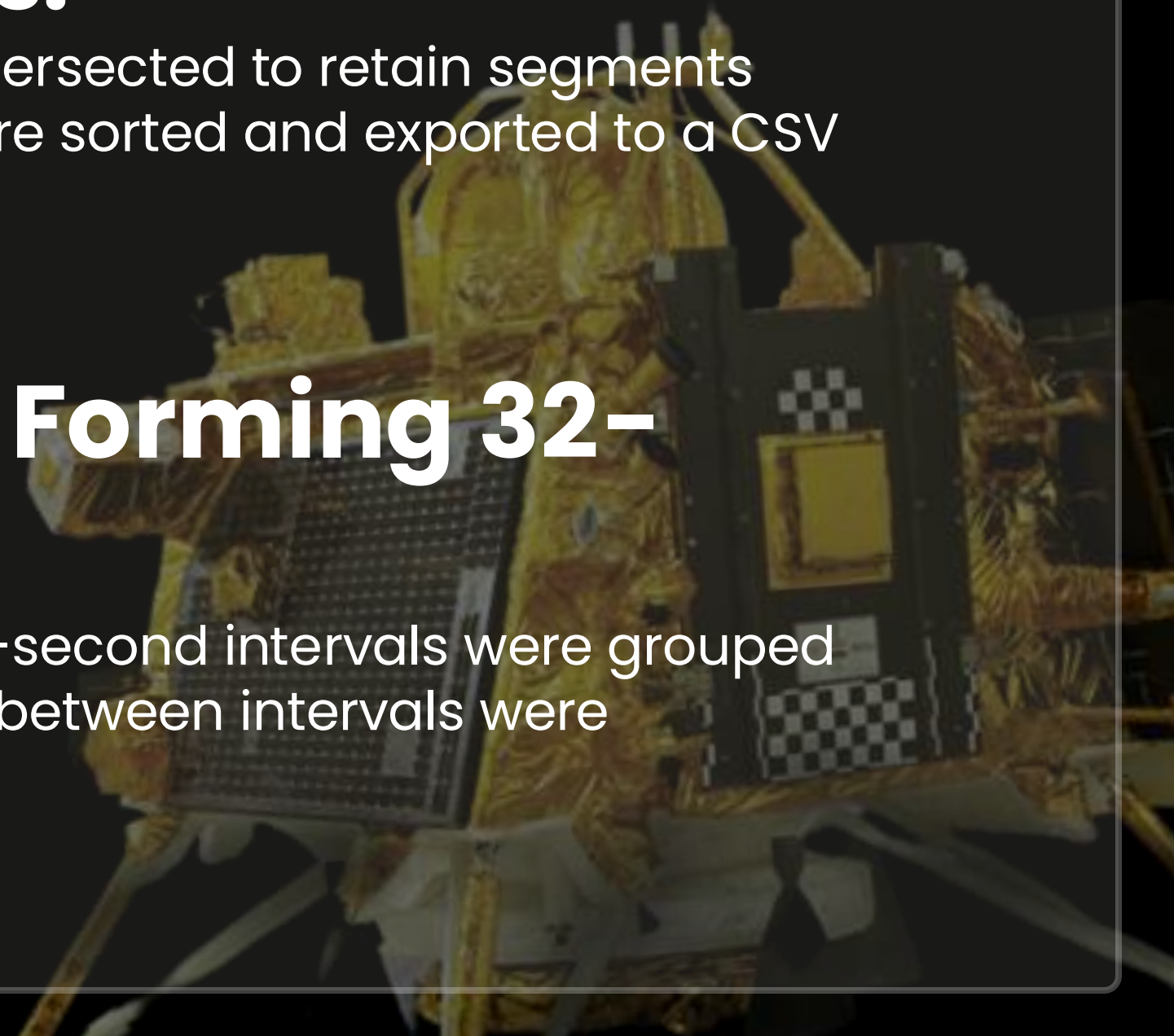
Raw GTI data from XSM was accessed daily for a specified month, retaining only quality segments. These intervals were converted from Mission Elapsed Time (MET) to Universal Coordinated Time (UTC) for standardization.

2. Filtering and Intersection of Intervals:

Two sets of intervals—8-second intervals and identified GTIs—were intersected to retain segments meeting both quality and duration criteria. The final valid intervals were sorted and exported to a CSV file for further use.

3. Ensuring Interval Continuity and Forming 32-Second Batches:

To meet the analysis requirement for longer durations, consecutive 8-second intervals were grouped into continuous 32-second segments. Only sets with exact continuity between intervals were accepted, ensuring valid and discrete 32-second data batches.



GEOTAIL

SPICE Kernels used

- **SPK (Spacecraft Planet Kernel):** Position, velocity and their respective timestamps.
- **CK (Camera Matrix Kernel):** Contains attitudinal data.
- **SCLK (Spacecraft Clock Kernel):** on-board clock calibration data required to perform a transformation between Ephemeris time (ET) and spacecraft on-board time (SCLK).
- **LSK (Leap Seconds Kernel):** Converts ET to UTC and vice-versa.
- **PCK (Planetary Constants Kernel):** Contains physical, dynamical and cartographic constants for target bodies.

Steps Followed:

- First, we obtained the SPICE dataset from the PRADAN dataset portal.
- We viewed the SPK kernel files in spact, a NASA utility and found that it contained various time intervals for when the satellite was in the night side of the orbit.
- Then during those intervals we generated background noise from the FITS files, whose solar angle was greater than 120 degrees.
- Now we simply applied thresholding at 7 counts to the data as we found out that during geotail, background data was spiking above average value.
- Then we stored those time intervals where the background data crossed the threshold.